



Problem Statement: AI-Driven Cash Flow Prediction & Risk Flagging System for Rural Micro Enterprises

Context

India has millions of rural micro enterprises, including SHGs, FPOs and individual entrepreneurs. These entities often lack formal credit histories, limiting their access to structured financial services.

However, the rapid adoption of digital payments (such as UPI), along with the availability of market intelligence and other proxy indicators, has created a rich data ecosystem reflecting real-time rural economic activity.

Despite this, these diverse datasets remain largely underutilised for generating predictive insights that can enhance financial stability and resilience.

Problem Statement

The current approach to monitoring the financial health of microenterprises is largely manual, limiting the ability of institutions and field officers to anticipate risks.

I. There is no integrated, data-driven system that combines financial data with digital transaction signals (e.g., UPI trends), market intelligence, and external factors to predict future cash flows.

II. As a result, early warning signs of financial stress are often missed, leading to delayed interventions, liquidity issues, and increased credit risk.

The challenge is to develop a predictive solution that enables proactive decision-making and timely support for such enterprises.

Objectives

- Develop an AI/ML-based model to predict cash flows over a **3–6-month horizon**.
- Integrate multi-source data, including:
 - **Financial records:** savings, loans, repayments, balances (if available)

- **Digital transaction proxies:** e.g., UPI trends and usage patterns
- **Market intelligence and historical data:** raw material and commodity prices, demand trends, long-term crop productivity data, climate data and seasonal patterns impacting income
- Design an early warning system to identify potential financial stress or risk in microenterprises.
- Provide simple, actionable insights for microenterprises and field officers to enable corrective actions.

Expected Deliverables

A working prototype (web/mobile interface) for micro enterprises and field officers.

For micro enterprises

- Data entry on savings/loans
- Income and expenses
- Risk alerts
- Actionable suggestions

For field officers

- List of micro enterprises and their risk status
- Detailed enterprise profiles
- Cash flow forecast view
- Risk panel

The solution should also include:

- Sector-specific risk for enterprises such as dairy, poultry, food processing, handicrafts, and rural retail.
- Climate and market risk integration, envisaging weather shocks, commodity price fluctuations, and local market disruptions into risk models.
- Dashboard for monitoring performance, including cash flow trends and key indicators.
- Risk alert system with clear visualisations (e.g., risk levels, early warning signals).

Assumptions / Constraints

- Use of mock or simulated datasets representing financials and proxy indicators.
- Should work offline / in low-network conditions.
- Data used (especially UPI or transaction proxies) should not rely on sensitive personal information.
- Multilingual support (optional).

Value Creation for NABARD

1. Enhanced credit flow to underserved rural enterprises

By generating reliable cash-flow forecasts and risk profiles, the platform can strengthen banks' credit appraisal capabilities and lead to improved access to formal credit for rural enterprises.

2. Credit-led Rural Development

The platform can act as a link between grant-based interventions and institutional finance by helping enterprises demonstrate repayment capacity, and thereby avail formal credit for business expansion, technology adoption etc. rather than relying on grants.

3. Creation of a Digital Public Good

The platform can act as a common digital infrastructure for enterprise profiling, cash-flow assessment and risk monitoring.

4. Better outcomes for beneficiaries

NABARD-supported beneficiaries can receive actionable insights into business performance and market risks for informed decision-making.